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CS 405/CS 505 – Data Mining

Final project: Mountains vs. Beaches Preferences

1. Introduction

In this data mining project, you will explore **how people prefer to spend their vacations**—whether they are drawn to **beaches or mountains**. The project involves building a **classification system** to predict individuals' vacation preferences based on a given dataset. Using the classifiers we have discussed in class, you will practice **developing, implementing, and validating models in Python**. Your task will be to apply three different classifiers to the [dataset](#) and **analyze the features** that influence these preferences. The goal is to predict whether a person prefers the **beach or mountain setting** for their vacation. Follow the four steps described in section 2) of the project guidelines to implement and evaluate the performance of the classifiers.

2. Tasks

Step 1. Choose three of the following classifiers to use them for the prediction of vacation preferences:

- Decision Trees
- Naïve Bayes
- Neural Networks
- K-Nearest neighbors
- Support Vector machine

The choice of three classifiers should be justified.

Step 2. Assess the chosen three classifier in Step 1). The principle is to compare between the three classifiers and to use the concept of model validation seen in the assignment 2.

The overfitting problem should be taken into consideration during the experiments. A section of experimental results must be added and detailed. You can support your arguments with curves and scores.

Step 3. Possibility to increase the scores. You have to check the possibility to increase the obtained scores in Step 2. Innovation in the experiment to improve the obtained scores will be an asset.

Step 4. A conclusion has to be added at the end. It should mention what was achieved in this final project, drawbacks, and the possibility of the improvement of the obtained results.

3. Submission

You must submit the *pdf* file of your report. In fact, it must contain detailed explanation of each step. In addition, you have to submit your *.py* or *.ipynb* if you prefer using *Jupyter Notebook*. **Please, do not submit your final projet in .zip or .rar files.** If you do a transformation to the dataset, you will have to mention it on your report and you will have to submit it.